

DUET-CKD: Dual-purpose fine-tuned LLMs for predictive robustness and human-centered explainability in chronic kidney disease

Abstract

Deployment of clinical prediction systems with explainable and user-friendly reasoning is important for their safe and trusted use in healthcare. Traditional machine learning (ML) models based on structured clinical data can achieve high predictive performance, but their reliance on technical post-hoc explainability methods often makes them inaccessible to non-technical users. Large language models (LLMs) have recently shown promise for generating human-friendly explanations. In this work, we employ fine-tuned LLMs (FT-LLMs) for a dual purpose, functioning both as a prediction model and as an explainability tool that generates user-friendly reasoning for chronic kidney disease (CKD). Specifically, we fine-tune a quantized, lightweight, general-purpose LLM on serialized clinical tabular data using parameter-efficient techniques, substantially reducing memory and computational demands. We evaluate the proposed DUET-CKD framework on three CKD datasets of varying scale (PKC, UCI, and SPN), benchmarking it against a random forest (RF) baseline across both predictive and explainability dimensions. The FT-LLM achieved classification performance comparable to RF on the smaller PKC (accuracy 0.95 for both) and UCI (accuracy 1.00 for both) datasets, while accuracy dropped on the large-scale SPN dataset (0.93 for the FT-LLM versus 0.99 for RF); the FT-LLM consistently achieved 100% recall across all three datasets, at the cost of a corresponding increase in false positives. These results show that traditional ML models remain more robust for large-scale tabular prediction, whereas the FT-LLM delivers comparable classification performance on small to medium-sized datasets while substantially improving the accessibility and usability of model explanations through natural language rationales. Overall, these findings reveal a trade-off between predictive robustness and human-centered explainability, suggesting that combining the complementary strengths of traditional ML and FT-LLMs may be necessary for designing clinical AI systems that are accurate, trustworthy, and suitable for real-world deployment.

Keywords: Large language models, Explainable AI, Parameter-efficient fine-tuning, Natural language rationales, Chronic kidney disease, Clinical prediction model, Machine learning

1 Introduction

Chronic Kidney Disease (CKD) is a progressive decline in kidney function that often goes unnoticed until the disease has advanced significantly [1]. It reduces the glomerular filtration rate, leaving patients reliant on dialysis therapy or a kidney transplant [2, 3]. CKD is a major global health issue, affecting nearly 10% of adults worldwide, more than 800 million individuals in total, and it burdens healthcare systems because of its prevalence and the high cost of treatment [4]. Early detection is of great significance: it improves patient survival relative to late diagnosis and reduces the burden on healthcare systems [5]. Traditional diagnostic techniques, primarily glomerular filtration rate (GFR) measurement and urine albumin tests, are useful for diagnosing and monitoring CKD but are less effective at predicting how the disease will progress or identifying the factors that most contribute to its onset [6].

Recent studies demonstrate that supervised machine learning (ML) and deep learning (DL) models trained on structured clinical data can predict CKD presence, stage, and progression with remarkable accuracy, often reaching 99% or higher [7–12]. Tree-based ML methods, including random forest (RF), extreme gradient boosting (XGBoost), and artificial neural networks (ANN), consistently demonstrate strong predictive performance across studies and emerge as top performers [13].

Despite these advances, most models still behave as black boxes that are difficult to interpret and provide little information about why a particular prediction is made [3]. Clinicians need not only accurate risk scores but also transparent, patient-specific explanations that facilitate decision making and help identify potential model errors or biases [14, 15]. In response, recent studies have increasingly integrated predictive modeling with explainable artificial intelligence (XAI) techniques [13, 16]. Methods such as Shapley additive explanations (SHAP), local interpretable model-agnostic explanations (LIME), and fuzzy rule-based systems are commonly used to highlight the contribution of individual features to each prediction [17].

XAI techniques offer a useful perspective on how features affect a model but generally produce technical attributions and plots [18, 19]. These outputs require expert knowledge to interpret and stop short of providing intuitive, human-friendly explanations of individual predictions, as observed by [20]. Rationalization, by contrast, seeks to explain model outputs in natural language, describing the reasoning behind a particular prediction [21, 22]. Because such rationales are written in a way that is easily understood, they remain accessible to non-technical users, which makes them particularly valuable in clinical settings where transparency and trust are essential [23].

LLMs have also emerged as powerful tools capable of generating human-like natural language responses to a wide range of medical queries, including those in the CKD domain [24], and researchers are now beginning to explore their performance on tabular data [25–27]. To address the need for user-friendly, justifiable explanations highlighted by earlier work [2, 21], we propose a methodology that draws on the natural language generation (NLG) capabilities of LLMs to produce intuitive, easy-to-follow rationales for CKD prediction, contributing to the broader integration of artificial intelligence (AI) into kidney disease care [28, 29].

Specifically, the objectives of this work are:

- (i) To propose a novel FT-LLM-based approach that generates case-specific natural language rationales explaining each CKD prediction;
- (ii) To benchmark FT-LLMs against traditional ML models for CKD prediction on clinical data.

The primary focus is to employ LLMs directly for classification and analyze their predictive performance and reasoning, an aspect that has not been explored in previous studies. The remainder of this paper is organized as follows: Section 2 reviews previous work, Section 3 describes the proposed methodology, Section 4 presents model training and evaluation, Section 5 presents the experimental results and discussion, and Section 6 concludes the paper.

2 Previous Work

ML and DL models are extensively used in healthcare for disease prediction and prognosis. In nephrology, and specifically for CKD, several studies have focused on CKD prediction and on techniques to improve model performance. Almost all of these studies have utilized supervised learning techniques on electronic health records (EHRs), laboratory test results, and patient medical histories. Khalil et al. [30] compared logistic regression (LR), k-nearest neighbor (KNN), and tree-based models such as decision trees (DT) and RF for CKD classification. The dataset used was collected from the Bab Al-Rayan Clinic Center and consisted of 100 patient records. KNN, LR, and RF achieved an accuracy of 95%, while the DT achieved 90% accuracy. In a second scenario, feature selection using recursive feature elimination (RFE) was applied to the dataset, and the selected features were used for model training. Under this setting, KNN achieved the highest accuracy of 100%, followed by LR at 95%, while DT and RF achieved 90% accuracy.

Rafi et al. [10] utilized a dataset containing 25 variables and reduced its dimensionality using principal component analysis (PCA). Multiple algorithms were evaluated, and the results showed that the XGBoost classifier outperformed all other models with an accuracy of 99.1%. In a separate study [31], ANNs were compared with ensemble methods such as XGBoost and RF. The dataset for this study was obtained from the University of California Irvine (UCI) ML Repository. The ANN achieved the highest accuracy of 97.5%, followed by XGBoost at 96.25% and RF at 96%, respectively.

Mandol et al. [32] evaluated four models, including KNN, RF, support vector machines (SVM), and naive bayes (NB), using the CKD dataset from the UCI ML Repository. SVM achieved the highest accuracy of 99.17%, followed by KNN and RF. NB performed the worst, with an accuracy of 63.33%, due to its assumption of feature independence. Another study [33] compared adaptive boosting (AdaBoost) classifier, categorical boosting (CatBoost), stochastic gradient boosting (SGB), gradient boosting classifier (GBC), XGBoost, and Extra Trees Classifier (ETC), in addition to previously mentioned models. ETC and XGBoost achieved the highest performance, both reaching an accuracy of 99.16%. These studies indicate that nearly all major supervised learning models have been explored for CKD prediction.

Recent literature also indicates that the use of XAI methods for simplifying CKD prediction models is increasing and proving useful in clinical practice. Studies based on XAI techniques such as SHAP and LIME have aimed to analyze feature importance at both the global model level and the level of individual patient predictions [6, 13, 21]. Jawad et al. [14] introduced partial dependence plots (PDP) and accumulated local effects (ALE) in addition to SHAP and LIME. PDPs provide a global view of how changes in one or two features, e.g., hemoglobin or serum creatinine, influence the predicted outcome, while ALE refines this view by accounting for correlations among features, yielding more reliable local explanations of how various biomarkers contribute to CKD risk. Huang et al. [34] combined SHAP with interpretability techniques based on Bayesian optimization, calibration curves, and macro-averaged fairness measures, extensions that enhanced the clinical credibility and dependability of the model’s forecasts. Collectively, this literature shows that XAI methods have increasingly become the primary means of making CKD prediction models more comprehensible, clinically valid, and reliable.

Beyond traditional interpretability, the emergence of LLMs has opened new possibilities for human-centered AI in healthcare. In recent years, researchers have begun exploring the application of LLMs to structured clinical data [35]. Frameworks such as TabLLM [25] and TabPFN [27] showed that LLMs can perform similarly to, or better than, traditional ML models on zero-shot or few-shot classification tasks involving tabular health records. Nevertheless, these same studies indicate that LLMs do not consistently outperform traditional ML models on tabular prediction, a finding also reported by [36]. Moreover, the majority of prior work focuses primarily on predictive accuracy; a significant gap remains in applying self-explanation prompting to generate case-specific, natural language clinical rationales for CKD prediction.

3 Methodology

This section presents the proposed framework in a stepwise manner. It begins with data collection and transformation, then proceeds to two parallel modeling pipelines: a conventional ML pipeline and an LLM-based pipeline. In the LLM pipeline, structured clinical data is converted to natural language through serialization, after which parameter-efficient fine-tuning is applied so that the model can be used to generate both predictions and explanations. The baseline ML model, in contrast, is trained directly on the tabular data to provide a reference level of performance. The two models are then evaluated using standard classification metrics, and their predictive performance and interpretability are compared. Figure 1 illustrates the complete workflow, including data transformation, training, fine-tuning, and evaluation, which is also formally described in Algorithm 1.

3.1 Data Collection

The study utilized three CKD-related datasets, each including both CKD-positive and CKD-negative patients, obtained from clinical centers and open-source data repositories. Using multiple datasets rather than a single source allowed us to assess whether

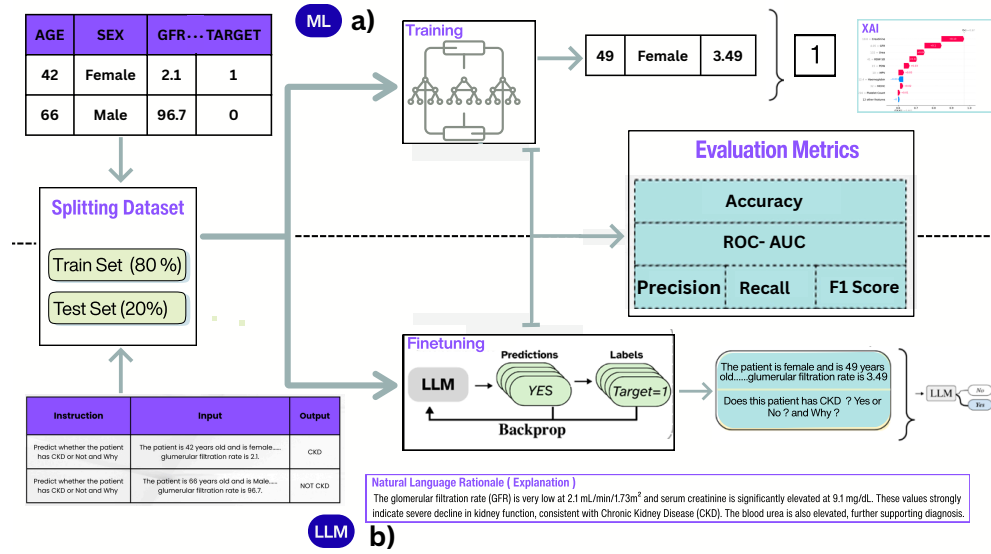


Fig. 1 Overview of the proposed DUET-CKD framework. (a) illustrates the ML pipeline using structured tabular data, while (b) depicts the LLM pipeline leveraging natural language inputs and rationales.

the proposed method generalizes across different data distributions, rather than reporting findings tied to one particular source. Together, the datasets comprise demographic attributes (e.g., age and gender), laboratory measurements (e.g., serum creatinine and blood urea), and clinical indicators (e.g., blood pressure and comorbidities) that are frequently used in CKD diagnosis.

Algorithm 1 DUET-CKD for Prediction and Natural Language Explanation

```
1: Input: CKD dataset  $D$ , pre-trained LLM  $M_\theta$ 
2: Output:
3:   Models  $M = \{M_{\theta'} \text{ (FT-LLM)}, RF_\phi \text{ (RF)}\}$ 
4:   Metrics  $V_{\text{eval}} = \{\text{Accuracy, Precision, Recall, F1-score, ROC-AUC}\}$ 
5:   Explainability  $E = \{R \text{ (NL Rationales)}, \text{XAI (SHAP)}\}$ 
6:
7: Step 1: Data Preprocessing
8: for each dataset  $D_i \in D$  do
9:   Handle missing values via column-wise imputation
10:  Remove outliers using IQR method: discard  $x \notin [Q_1 - 1.5 \cdot \text{IQR}, Q_3 + 1.5 \cdot \text{IQR}]$ 
11:  if ML pipeline then
12:    Encode categorical features; normalize numerical attributes
13:    Split  $D_i \rightarrow D_{\text{train}}$  (80%),  $D_{\text{test}}$  (20%)
14:  else (LLM pipeline)
15:    Define serialization template  $T_s$  in instruction-tuning format
16:    Convert each record  $x_i \rightarrow$  natural language:  $s_i = \text{Serialize}(x_i, T_s)$ 
17:    Split  $S \rightarrow S_{\text{train}}$  (80%),  $S_{\text{test}}$  (20%)
18:  end if
19: end for
20:
21: Step 2: Baseline ML Training (RF)
22: Initialize RF classifier  $RF_\phi$ 
23:  $RF_\phi \leftarrow \text{Train}(RF_\phi, D_{\text{train}})$ 
24:  $\hat{y}_{RF} \leftarrow RF_\phi.\text{predict}(D_{\text{test}})$ 
25: Store performance metrics for benchmarking
26:
27: Step 3: Parameter-Efficient FT (QLoRA)
28:  $M_q \leftarrow \text{LoadQuantized}(M_\theta, \text{NF4})$  // load base model in 4-bit NF4
29:  $M_{\theta'} \leftarrow \text{Train}(M_q, S_{\text{train}})$  // FT via LoRA adapters, 8-bit AdamW
30:  $\hat{y}_{LLM} \leftarrow M_{\theta'}.\text{predict}(S_{\text{test}})$ 
31:
32: Step 4: Evaluation and Reasoning Comparison
33: Evaluate  $M_{\theta'}$  and  $RF_\phi$  on test sets  $S_{\text{test}}, D_{\text{test}}$ 
34: Compute confusion matrix:  $\{TP, TN, FP, FN\}$ 
35: Derive  $V_{\text{eval}} = \{\text{Accuracy, Precision, Recall, F1-score, ROC-AUC}\}$ 
36:  $\text{XAI} \leftarrow \text{SHAP}(RF_\phi, D_{\text{test}})$  // feature-level explanations
37:  $R \leftarrow M_{\theta'}.\text{explain}(S_{\text{test}})$  // NL rationales via self-explanation prompting
38: return  $M, V_{\text{eval}}, E = \{R, \text{XAI}\}$ 
```

The datasets used in this study are described below. The Pakistan Kidney Center (PKC) dataset is a location-specific dataset collected at the Pakistan Kidney Center in Abbottabad, containing real-world patient records [37]. The UCI CKD dataset was accessed via the UCI ML Repository and is freely available [38], allowing direct comparison with previous studies. The Spanish CKD dataset (SPN) was obtained from the Kaggle platform [39]; it is referred to as “Spanish” because it was originally recorded in Spanish and later translated into English, and it is publicly accessible as the *insuficiencia renal crónica* dataset on Kaggle. These three datasets differ substantially in size, enabling evaluation across different data regimes: PKC and UCI are small-scale clinical datasets with several hundred records each, while SPN is a large-scale dataset with more than 15,000 records. This variation allows model performance to be analyzed in both low-data and high-data scenarios. Dataset characteristics are summarized in Table 1, and the class distribution of CKD and non-CKD instances across the three datasets is presented in Table 3.

Table 1 Summary of dataset characteristics used in the study.

Dataset Name	Total Features	Categorical	Numerical	Shape (rows, cols)
PKC	22	2	20	398, 22
UCI	25	11	14	400, 25
SPN	20	6	14	15050, 20

3.2 Data Preprocessing

The datasets were checked for null values. The UCI and SPN datasets both contained null values, while the PKC dataset was clean with no null values. Row removal and imputation techniques were used to address the missing values in the affected columns. Outliers in the datasets were then checked. The SPN dataset contained significant outliers; therefore, the Interquartile Range (IQR) method was used to identify and remove these anomalies. For each numerical column x , the first (Q_1) and third (Q_3) quartiles were computed, and the IQR was defined in Eq. 1:

$$IQR = Q_3 - Q_1 \quad (1)$$

The lower and upper bounds were set as shown in Eq. 2:

$$L = Q_1 - 1.5 \times IQR \quad \text{and} \quad U = Q_3 + 1.5 \times IQR \quad (2)$$

Any observation where $x < L$ or $x > U$ was treated as an outlier and removed from the dataset prior to model training. Following the implementation of these preprocessing steps, Table 2 illustrates how the datasets were transformed compared to the initial statistics in Table 1.

This class distribution is important for understanding potential biases and for selecting appropriate evaluation metrics during model training and testing.

Table 2 Dataset characteristics after preprocessing.

Dataset Name	Total Features	Categorical	Numerical	Shape (rows, cols)
PKC	22	2	20	398,22
UCI	25	11	14	273,25
SPN	20	6	14	10987,20

Table 3 Class distribution of CKD and Not CKD cases across the three datasets.

Dataset Name	CKD	Not CKD	Total	CKD Proportion
PKC	241	157	398	60.6%
UCI	128	145	273	46.9%
SPN	5842	5145	10987	53.2%

3.3 Data Serialization for LLMs

We use LLMs to perform predictions on clinical data; however, unlike traditional ML models, they cannot directly process tabular data. To enable LLM processing, the table must be transformed into natural language text. We used the Text GPT-3 method to serialize the tabular data into natural language text, as described in [25].

Figure 2 illustrates the transformation of each tabular patient record into an LLM-readable textual sequence via serialization, following an instruction-tuning format (instruction, input, and output) used for model training and inference.

This serialized format is used for model training and inference, enabling LLMs to generate predictions and rationales from clinical descriptions. The data format shown in Figure 1 (b) is derived using the serialization method depicted here.

4 Model Training and Evaluation

This section presents the predictive modeling strategies along with the evaluation methods adopted in this study. We used a traditional ML model training approach and LLM fine-tuning to compare the results, as depicted in Figure 1 (a) and (b). Both models were trained and evaluated on identical datasets using an 80/20 train-test split to ensure a fair and consistent benchmarking protocol.

4.1 ML Model as Baseline

In this work, we select RF as the main ML baseline to compare against the LLM-based approach. RF was selected because it is widely used for tabular classification and is known to provide strong average results across a wide range of datasets [2]. RF combines predictions from multiple decision trees to reduce variance and improve generalization, making it especially suitable for structured clinical data such as CKD patient records. Its popularity in medical ML research also supports its use as a stable benchmark against which the LLM results can be compared. Ensemble-based

approaches such as XGBoost and Gradient Boosting have also proven competitive in CKD research, but RF was chosen as the primary baseline for two reasons. First, it is among the most popular and consistently high-performing models on clinical tabular data, making it a credible and repeatable benchmark across datasets of different scales. Second, the ensemble-averaging mechanism of RF makes it inherently resistant to overfitting, and its parallel structure yields more stable and reproducible results than sequential boosting models [40, 41]. More importantly for this work, RF is inherently compatible with post-hoc explainability techniques such as SHAP [3, 15, 21], which allows a direct and meaningful comparison between technical feature-attribution explanations and the natural language rationales produced by the FT-LLM, the primary explainability goal of this work [19, 23].

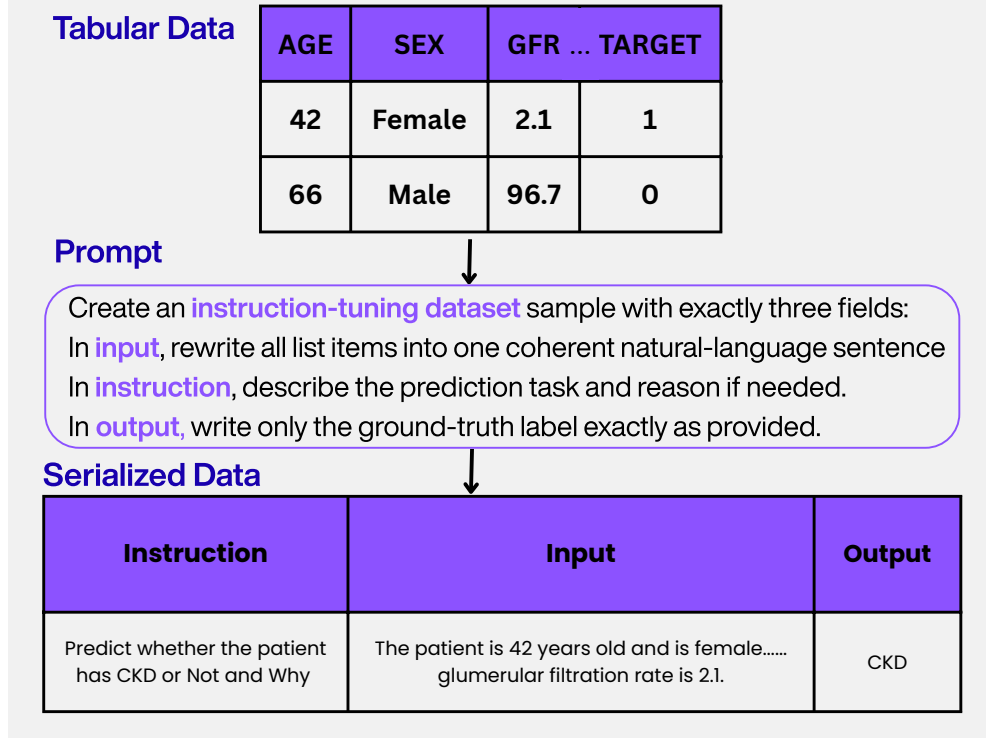


Fig. 2 Transformation of tabular patient records into LLM-readable textual sequences via serialization.

4.2 Large Language Model Fine-Tuning on Serialized Data

A pre-trained LLM was employed and fine-tuned for CKD prediction and explanation. Specifically, the **Gemma-3N-E4B-it** model was selected, an instruction-tuned variant of Gemma 3N with an effective parameter count of 4 billion (E4B), loaded in 4-bit quantization via Unsloth to reduce memory consumption. Training was conducted on

a single NVIDIA T4 GPU (16 GB VRAM) provided by the free tier of Google Colab. This model was chosen for its efficient architecture and strong performance relative to its computational footprint, making it well suited to the limited compute environment.

4.2.1 Parameter-Efficient Fine-Tuning

To reduce computational cost and mitigate overfitting, Parameter-Efficient Fine-Tuning (PEFT) [42] was employed using Low-Rank Adaptation (LoRA) [43]. LoRA works by keeping base model parameters frozen and introducing only a small set of trainable low-rank adapter matrices into the attention and feed-forward layers of the model. Additionally, 4-bit quantization was applied to the base model weights (QLoRA) [44], significantly reducing memory and computational requirements with negligible impact on model performance. The model was fine-tuned using supervised instruction tuning with gradient accumulation to enable a larger batch size.

During supervised fine-tuning, we configured the training loop with a maximum of 60 steps for PKC and UCI, and 100 steps for SPN, using different step counts to account for the differing dataset sizes. To support larger effective batch sizes without exceeding memory limits, we applied gradient accumulation with 4 steps. For optimization, we selected 8-bit AdamW with a weight decay of 0.01 to regularize the model and help mitigate overfitting. Within LoRA, we set the rank (r) to 8 and LoRA alpha to 8, which controls the expressiveness and scaling of the low-rank adapter matrices inserted into the attention and feed-forward layers. These hyperparameters, in conjunction with QLoRA quantization and frozen base weights, enabled efficient fine-tuning that preserved performance while greatly reducing compute and memory demands.

4.2.2 Evaluation Metrics

Predictive performance is quantified using the mathematical formulations in Table 4 [45], where TP , TN , FP , and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively.

5 Results and Discussion

This section presents the experimental results obtained after benchmarking the models against three different datasets. We evaluate both predictive performance and the quality of generated explainability by each model, highlighting the strengths and limitations of each approach.

5.1 Predictive Performance

The predictive performance of the FT-LLM and the RF baseline was evaluated on the test sets of the PKC, UCI, and SPN datasets using the metrics presented in Table 4. Results are summarized in Table 5.

Across the PKC and UCI datasets, the FT-LLM achieved performance comparable to the traditional ML baseline. Notably, both models achieved perfect classification on the UCI dataset, which is consistent with prior studies on this dataset [32, 33]

Table 4 Evaluation metrics for CKD prediction.

Metric	Formula	Description
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Overall proportion of correctly classified CKD and non-CKD cases.
Precision	$\frac{TP}{TP + FP}$	Measures the proportion of predicted CKD cases that are truly CKD, reflecting the model’s ability to avoid false positives.
Recall	$\frac{TP}{TP + FN}$	Measures the proportion of actual CKD cases correctly identified by the model (sensitivity).
F1-Score	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	Harmonic mean of precision and recall, providing a balanced measure of classification performance.
ROC-AUC	$\int_0^1 \text{TPR}(t) d[\text{FPR}(t)]$	Measures the model’s ability to distinguish between CKD and non-CKD cases across all decision thresholds.

Table 5 Predictive performance of the RF and FT-LLM models on the PKC, UCI, and SPN CKD datasets.

Dataset	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
PKC	RF	0.95	0.96	0.96	0.96	0.95
	FT-LLM	0.95	0.91	1.00	0.95	0.95
UCI	RF	1.00	1.00	1.00	1.00	1.00
	FT-LLM	1.00	1.00	1.00	1.00	1.00
SPN	RF	0.99	0.99	0.99	0.99	0.99
	FT-LLM	0.93	0.89	1.00	0.94	0.93

and attributable to its small size, clean structure, and the high discriminative power of its clinical biomarkers following preprocessing. On the PKC dataset, the FT-LLM achieved competitive performance despite being fine-tuned on a small training set, suggesting that serialized clinical narratives carry sufficient discriminative signal for the model to learn meaningful decision boundaries in low-data regimes.

A performance decline was observed on the larger SPN dataset, consistent with prior studies [25, 36] showing that LLMs perform well in few-shot or low-data settings but do not outperform traditional ML models on large-scale tabular data. This degradation can be attributed to two factors: first, the serialization process may not fully preserve the complex feature interactions that tree-based models capture naturally through recursive partitioning; second, the limited fine-tuning steps (100 for SPN) and constrained compute environment may have been insufficient for the model to generalize across the larger and more varied SPN cohort. In contrast, RF maintained stable and consistently high performance across all three datasets, confirming its robustness as a tabular classification baseline regardless of dataset scale.

This behavior is further confirmed by the confusion matrices shown in Figure 3. On the PKC dataset, the FT-LLM produced a higher number of false positives and zero

false negatives, as illustrated in Figure 3(d), while successfully identifying all CKD-positive cases. This pattern becomes more evident on the large-scale SPN dataset, where the FT-LLM incorrectly labeled 147 non-CKD instances as CKD (Figure 3(f)), whereas the RF model demonstrated more robust pattern extraction with only one false positive and one false negative (Figure 3(c)).

This behavior indicates that the FT-LLM prioritizes recall over precision in the evaluated setting (Table 5), consistently achieving 100% recall across all three datasets.

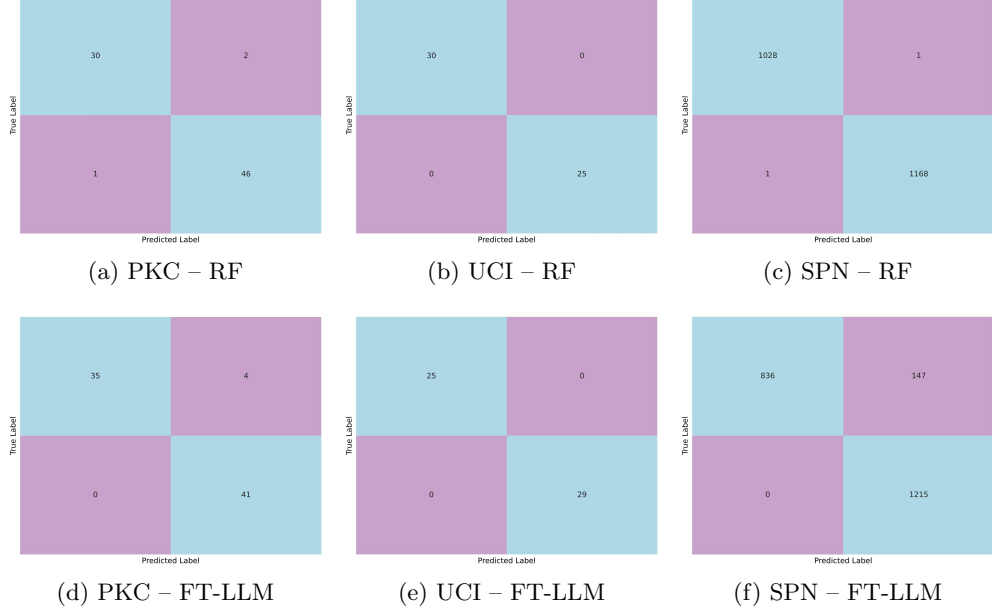


Fig. 3 Confusion matrices for CKD prediction using the RF and FT-LLM models across the PKC, UCI, and SPN datasets: (a–c) RF; (d–f) FT-LLM.

These results are consistent with prior literature [25, 36] indicating that LLMs are effective on small to medium sized tabular data but fall behind traditional ML models on larger structured data. The FT-LLM consistently favored recall over precision in our experiments, achieving 100% recall on all datasets. Nonetheless, this focus on recall amplifies false positives, in which healthy patients are falsely flagged as having CKD. This is a challenge in clinical practice, since it can lead to unnecessary follow-up investigations, increased workload for clinicians, and decreased confidence in automated tools; as a result, the FT-LLM is less practical than more balanced traditional models in this setting.

5.2 Explainability Comparison

We use a framework focused on accessibility and usability to compare the explainability of both models, as summarized in Table 6. This framework evaluates how effectively

each model communicates its predictions to end users. Figure 4 provides a side-by-side comparison of how each model justifies its predictions in two scenarios: a positive CKD case and a negative case.

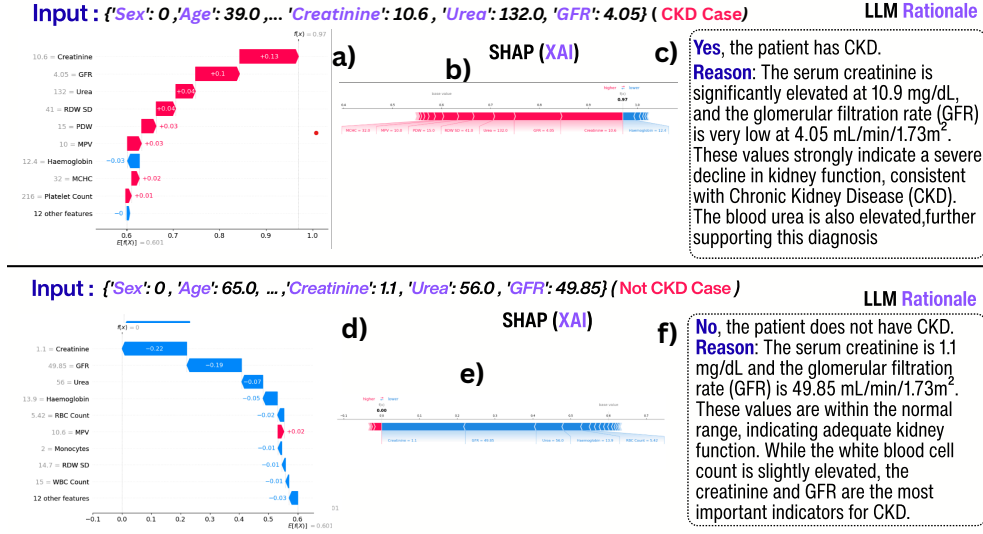


Fig. 4 Comparison of Technical XAI (SHAP) vs. LLM Natural Language Rationales.

For traditional ML, Figure 4(a) and 4(d) represent SHAP waterfall plots, which break down the final prediction into individual feature contributions, while Figure 4(b) and 4(e) display SHAP force plots, showing the mathematical push of features toward a specific outcome. In contrast, Figure 4(c) and 4(f) present the LLM natural language rationales, which provide a structured narrative explaining the diagnosis in plain text.

Using the proposed framework and the findings shown in Figure 4, we conclude that the LLM offers the more effective approach for communicating predictions in clinical settings. SHAP relies on abstract visualizations and mathematical associations, whereas the LLM offers contextual reasoning that translates raw clinical values directly into a diagnostic narrative. SHAP plots are primarily designed for developers and data scientists, whereas LLM rationales are designed for clinicians and real-world diagnostic decision support. As a result, the LLM requires little effort to understand: a practitioner can verify the reasoning in seconds, whereas interpreting a force plot requires greater cognitive effort to translate statistics into health outcomes. Thus, based on the criteria in Table 6 and the visual comparison in Figure 4, the FT-LLM emerges as the more effective explainability solution for clinical use.

This paper examined the viability of fine-tuning LLMs for CKD classification and explanation, comparing them to an RF baseline across several datasets. The results reveal a clear trade-off between predictive performance and explainability, underscoring that the suitability of either method depends on the clinical purpose it is intended to serve.

Table 6 Comparison of explainability frameworks.

Feature	SHAP/LIME (Traditional ML)	(Traditional ML)	Natural Language Rationale (LLM)	Clinical Focus	References
Output Format	Technical graphs and numerical weights		Structured narrative sentences	LLM	Figure 4
Target Audience	ML experts / data scientists		Clinicians / non-technical users	LLM	Figure 4
Understanding Effort	High (requires technical literacy)		Low (immediate, intuitive reading)	LLM	[21]
Granular Quantification	Provides exact numerical SHAP values per feature		Describes feature importance qualitatively	SHAP	[3, 14, 15]
Reproducibility & Consistency	Deterministic output (same input produces the same output)		Non-deterministic (outputs may vary for the same input)	SHAP	[24]

In terms of predictive accuracy, traditional ML models remain better suited to large, structured clinical datasets. RF maintained stable performance across all three datasets, including the large SPN cohort, whereas the FT-LLM matched this performance only on the small to medium-sized datasets. The decline in performance on larger data supports prior research showing that LLMs perform well in few-shot or low-data settings but do not consistently outperform classical models on large-scale tabular prediction. LLMs are also inherently ill-suited to regression-based objectives, such as continuous risk scoring or predicting CKD progression, where numerical accuracy and temporal consistency are important.

Despite these shortcomings, LLMs offer a tangible benefit in explainability. SHAP-based explanations provide mathematically sound feature attributions, but interpreting them requires a level of technical expertise that limits their practicality in everyday clinical use. The natural language explanations generated by the LLM, by contrast, transform raw clinical values into coherent, diagnosis-oriented narratives that resemble how clinicians reason, substantially reducing cognitive load and allowing non-technical users to verify predictions quickly.

That said, several limitations should be noted. First, LLM-generated explanations may not necessarily remain faithful to the underlying prediction process, since they are based on learned language patterns rather than the model’s internal reasoning, creating a risk of hallucinated or misleading justifications. Second, LLM performance is sensitive to the quality and design of the serialization step, which may not fully capture the intricate feature interactions present in tabular data. Finally, the relatively small size of the fine-tuned model, combined with limited computational resources, may have constrained its ability to generalize, particularly on large-scale datasets.

These findings reveal a trade-off that is, in practice, difficult to avoid: using LLMs as prediction models comes at the cost of a modest reduction in predictive capability in exchange for improved usability and interpretability. This trade-off can be worthwhile in clinical decision support systems, where transparency, trust, and rapid comprehension are all important. Consequently, LLMs are best viewed as complementary tools

rather than substitutes for traditional ML models, given their strength in providing user-centered explanations.

Importantly, this trade-off need not be framed as a binary choice between the two approaches. A hybrid architecture, in which a traditional ML model performs the prediction and an LLM generates the corresponding human-readable explanation, offers an alternative that combines strong predictive performance with improved interpretability and accessibility for clinicians.

Such a hybrid system could be implemented in daily clinical practice: ML model outputs would feed into the decision support workflow, with LLM-generated explanations reviewed promptly by clinicians. This human-in-the-loop approach may build trust, support informed decision-making, and reduce the cognitive load of interpreting complex model outputs.

Overall, this work underscores the importance of evaluating explainability from a human-centered perspective. Precise predictions remain better handled by traditional models, but LLMs surpass conventional explainability frameworks in providing intuitive, clinician-friendly reasoning, making them valuable components of the interpretable clinical AI systems still to be developed.

6 Conclusion

This paper proposes the DUET-CKD framework for predicting CKD using an FT-LLM and an ML model. The experimental results indicate a clear difference in the relative advantages of the two approaches. Whereas RF exhibits both predictive consistency and strength on datasets ranging from small to large, the FT-LLM offers user-friendly and highly intuitive natural language reasoning behind each prediction, which can greatly aid interpretability and usability. These results show that neither approach alone is sufficient to address the needs of practical prediction systems. Instead, a hybrid solution is needed, one that takes advantage of the strengths of ML models to make predictions that are accurate and reliable, and the strengths of LLMs to transform these predictions into human-friendly, clinically relevant explanations. Such a combination can help bridge the existing divide between high-performing predictive analytics and more accessible, trust-building explanations, making AI systems more practical for healthcare settings.

Future work will focus on designing hybrid architectures in which ML models serve as the primary predictor and LLMs serve as the explanation generator, preserving both accuracy and interpretability. Further studies are also needed to improve the fidelity of LLM-generated rationales, grounding them in the model’s actual behavior rather than plausible-sounding text generation. Research on more sophisticated serialization techniques, larger and more diverse clinical datasets, and real-time clinical validation will also be important steps toward deployment.

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Declarations

Ethics approval. Not applicable.

Informed consent. Not applicable.

Research involving human and animal rights. This research did not involve any human or animal subjects, as it relied on publicly available datasets.

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